

Automatic Daily Meal REMINDER SYSTEM

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A structured meal plan consistently supports a healthy lifestyle, regulating body weight and reducing the risk of diabetes and cardiovascular diseases.

Maintaining regular meal habits without external reminders can be challenging for students in rental accommodations. Skipped meals and irregular eating patterns often lead to health concerns and worrying for their families.

An automated system addresses this by providing timely meal reminders and notifying parents for added reassurance. Built around an Arduino-based setup, it ensures

adherence to a structured routine.

Table 1 lists the required components for the system, while Fig. 1 presents the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the automatic daily meal reminder system. It is built around the Arduino Uno (MOD1), RTC3231 (MOD2), SIM800L GSM module (MOD3), 16x2 LCD I2C module (MOD4), a buzzer, an LED, and a few other components.

An I2C display has been selected due to its ease of use, requiring only two wires for connection (SDA and SCL). This simplifies both the circuit design and the code implementation.

The system consists of an Arduino interfaced with an LCD, SIM900A/800L GSM module, RTC

module, buzzer, LED, and push-button switches. It simplifies adherence to a daily meal schedule by providing automatic reminders at predefined times. The system sends alerts and displays meal reminders on the LCD panel while also notifying parents via SMS regarding whether the student has followed the meal plan.

The system is placed near the student's study table or bed. It displays the current time and date using the

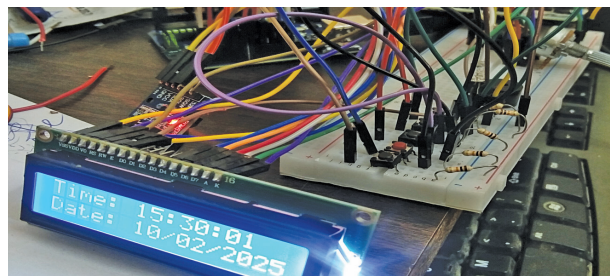


Fig. 1: Author's prototype

BILL OF MATERIALS

Name	Quantity
Arduino Uno (MOD1)	1
RTC3231 (MOD2)	1
SIM900A/SIM800L GSM module (MOD3)	1
16x2 LCD I2C module (MOD4)	1
Buzzer (B1)	1
LED (D1)	1
330Ω resistor (R1)	1
12V, 2A adaptor	1
Push-to-on switches (S1-S5)	5

Note: Here, the author has used the SIM900A module.

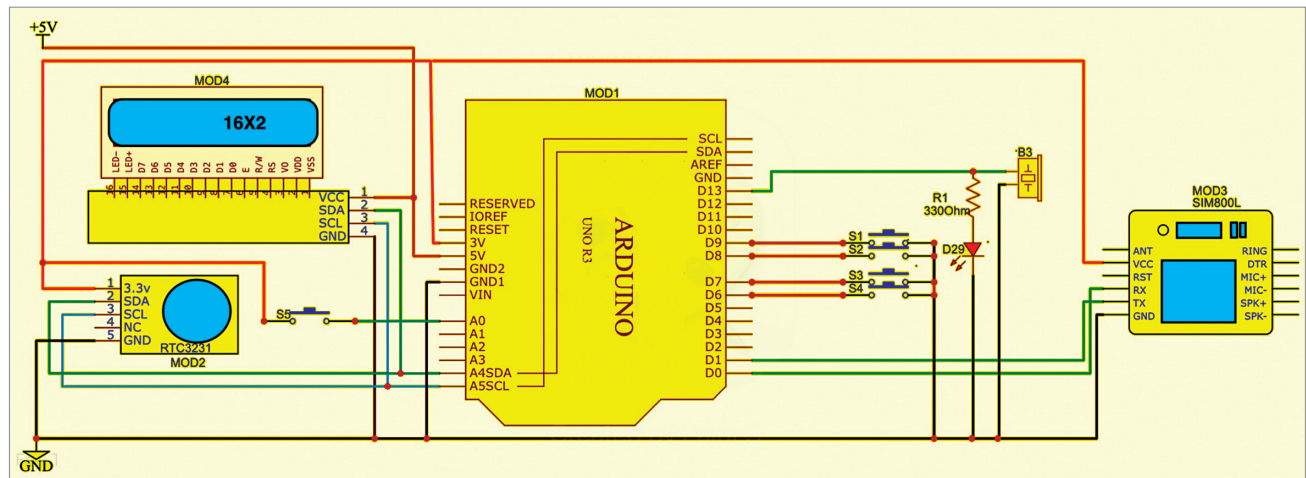


Fig. 2: Circuit diagram